



Buildpacks for Ops Teams

Photo by [Michele Bitetto](#) on [Unsplash](#)

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Slides



<https://github.com/dmikusa/buildpacks-for-ops>



Why are we here today?

1. A Quick primer
2. Problems Solved
3. Why Ops Teams Should Care?



What are Cloud Native Buildpacks?

What they are?

- Source Code => OCI
- Standard & specification
- Support Multiple Languages
- Integrated into Platforms

What they're not?

- Total replacement for Dockerfiles
- Replacement for CI/CD
- Dev-only Tool
- An implementation



What are Cloud Native Buildpacks?

Technically:

- A build + detect executable
- A metadata config file called `buildpack.toml`
- Bundled as an OCI image

• Current Layer Contents			
Permission	UID:GID	Size	Filetree
-----	0:0	12 MB	cnb
-----	0:0	12 MB	└─ buildpacks
drwxr-xr-x	0:0	12 MB	└─┬─ paketo-buildpacks_bellsoft-liberica
drwxr-xr-x	0:0	12 MB	└─┬─┬─ 11.5.4
-rw-r--r--	0:0	11 kB	└─┬─┬─┬─ LICENSE
-rw-r--r--	0:0	493 B	└─┬─┬─┬─ NOTICE
-rw-r--r--	0:0	29 kB	└─┬─┬─┬─ README.md
drwxr-xr-x	0:0	12 MB	└─┬─┬─┬─ bin
-rwxr-xr-x	0:0	0 B	└─┬─┬─┬─┬─ build → main
-rwxr-xr-x	0:0	0 B	└─┬─┬─┬─┬─ detect → main
-rwxr-xr-x	0:0	5.0 MB	└─┬─┬─┬─┬─ helper
-rwxr-xr-x	0:0	7.4 MB	└─┬─┬─┬─┬─ main
-rw-r--r--	0:0	28 kB	└─┬─┬─┬─┬─ buildpack.toml

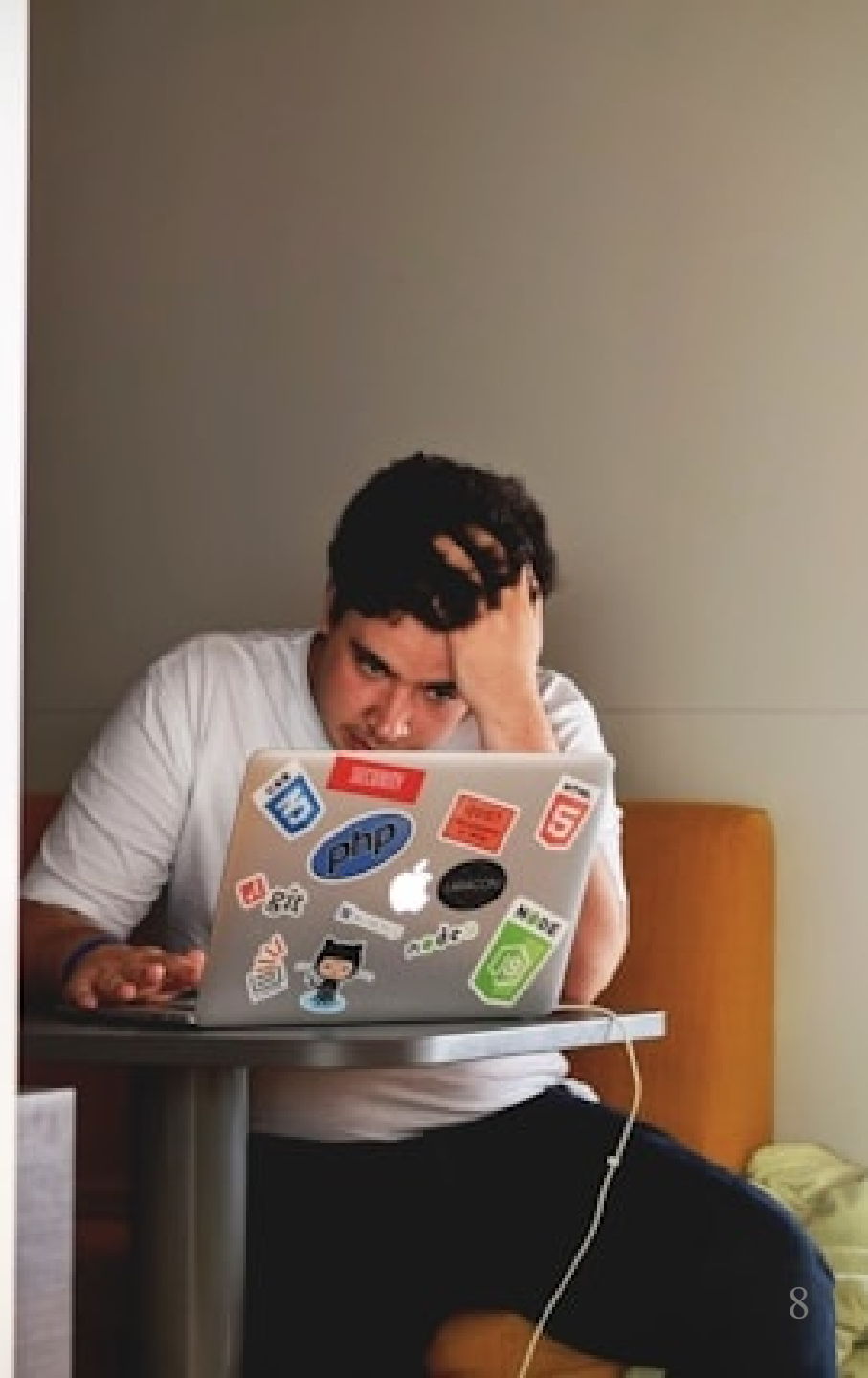
Using Buildpacks

- Install pack
- Run `pack build`

```
====> ANALYZING
[analyzer] Image with name "apps/nginx" not found
====> DETECTING
[detector] target distro name/version labels not found, reading /etc/os-release file
[detector] paketo-buildpacks/nginx 1.0.14
====> RESTORING
[restorer] No run metadata found at path "/cnb/run.toml"
====> BUILDING
[builder] target distro name/version labels not found, reading /etc/os-release file
[builder] Paketo Buildpack for Nginx Server 1.0.14
[builder] Resolving Nginx Server version
[builder] ...
[builder] Generating SBOM for /layers/paketo-buildpacks/nginx/nginx
[builder] Completed in 0s
[builder]
====> EXPORTING
[exporter] Adding layer 'paketo-buildpacks/nginx:nginx'
[exporter] Adding layer 'buildpacksio/lifecycle:launch.sbom'
[exporter] Added 1/1 app layer(s)
[exporter] Adding layer 'buildpacksio/lifecycle:launcher'
[exporter] Adding layer 'buildpacksio/lifecycle:config'
[exporter] Adding layer 'buildpacksio/lifecycle:process-types'
[exporter] Adding label 'io.buildpacks.lifecycle.metadata'
[exporter] Adding label 'io.buildpacks.build.metadata'
[exporter] Adding label 'io.buildpacks.project.metadata'
[exporter] Adding label 'io.buildpacks.exec-env'
[exporter] Setting default process type 'web'
[exporter] Saving apps/nginx...
[exporter] *** Images (9713097ce232):
[exporter] apps/nginx
Successfully built image apps/nginx
```

Problems Solved: Dockerfiles

- Dockerfile copy & paste
- Dockerfile sprawl



Problems Solved: Best Practices

- Base image too big
- Base image too small
- Base image not maintained
- Not your golden base image
- Forgets to use multi-stage builds
- Implements multi-stage incorrectly
- Bad layer ordering (cache invalidation)
- Bad layer breakdown (decreases sharing)
- Runs the app as root
- No PID1 handler
- Forget company CA certs
- exec vs shell to start process





Why Should Ops Care?

Photo by [Cytonn Photography](#) on [Unsplash](#)

How can Buildpacks Help Ops

1. Ops gets control



How can Buildpacks Help Ops

1. Ops gets control

2. Know what's running

How can Buildpacks Help Ops

1. Ops gets control
2. Know what's running

3. Building a Platform

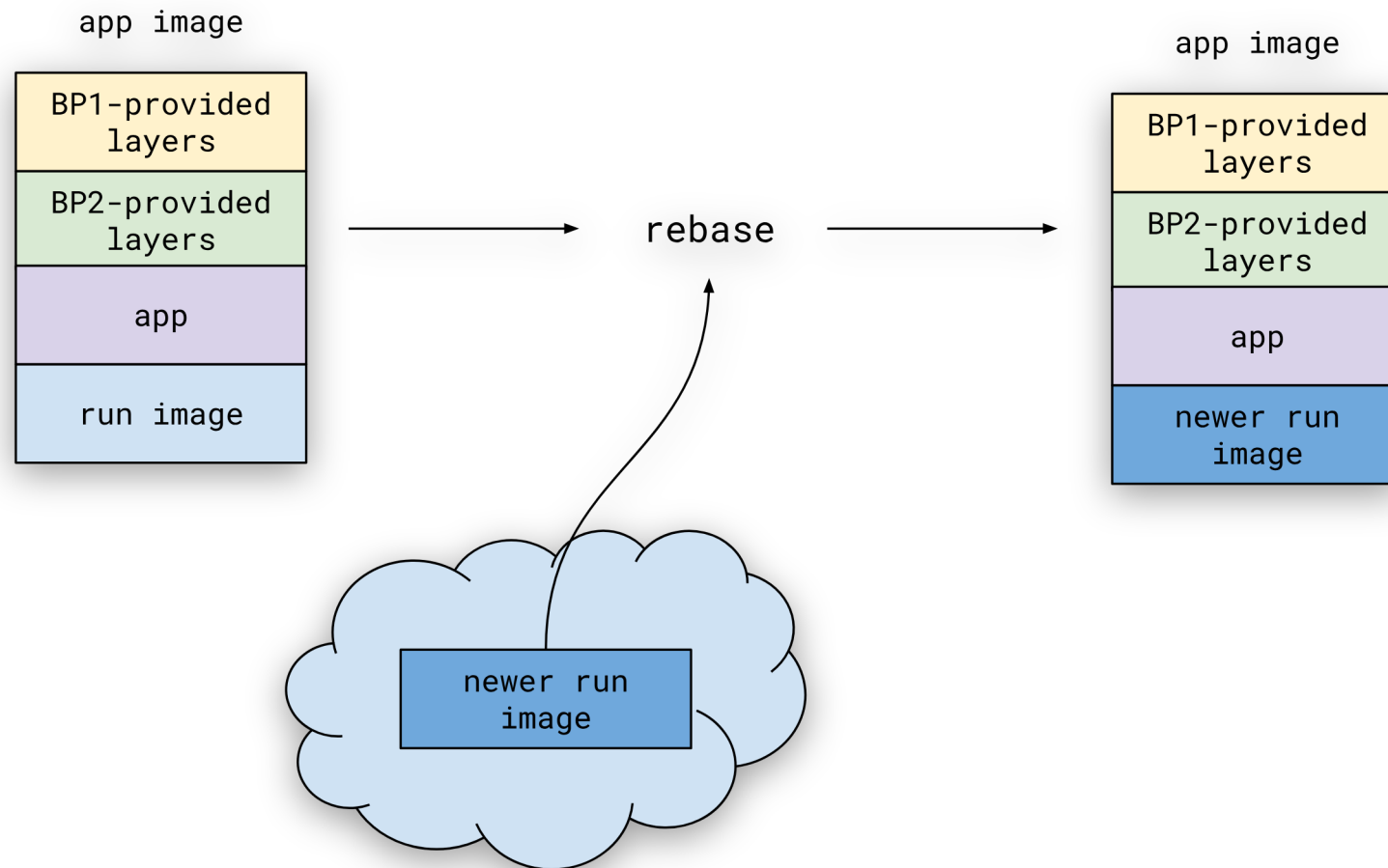




How can Buildpacks Help Ops

1. Ops gets control
2. Know what's running
3. Building a Platform
- 4. Fast rebuilds & rebase support**

Rebase Explained



How can Buildpacks Help Ops

1. Ops gets control
2. Know what's running
3. Building a Platform
4. Fast rebuilds & rebase support
- 5. Buildpacks are multi-language & multi-arch**



Levels of Customization

None

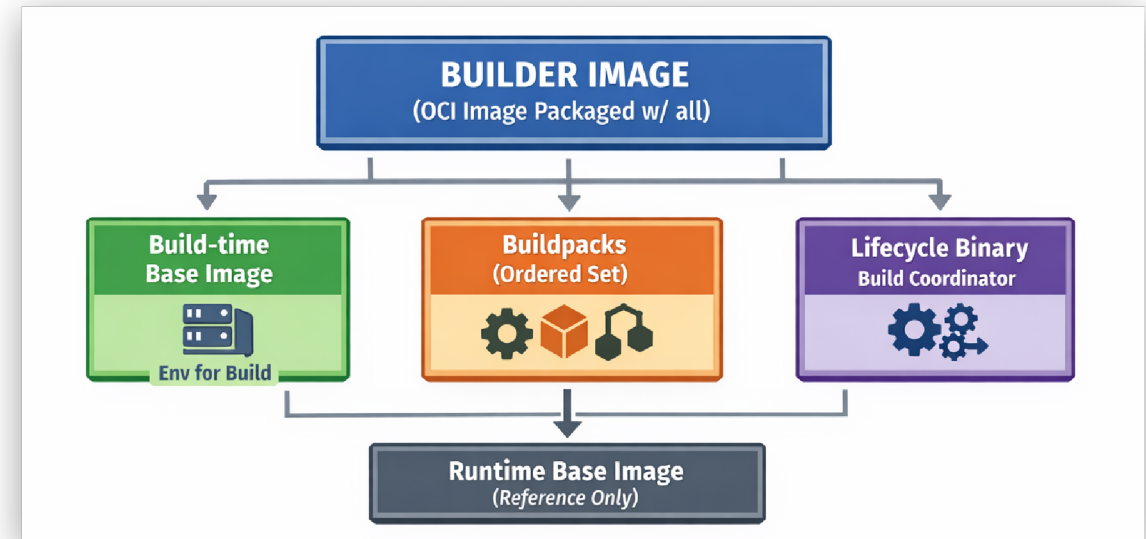
- Base images: Ubuntu Jammy, Ubuntu Noble, or UBI 8-10
- Four run image sizes: static, tiny, base & full
- Languages: Java, GraalVM, Scala, Kotlin, Clojure, Go, .NET, Node.js, Python, Ruby, Rust, PHP & web servers
- Buildpack-less builders too



Levels of Customization

Custom Builders

- Build image
- Run image
- Set of buildpacks
- Optional custom env variables



Levels of Customization

Custom Paketo Run Image

- Add more packages
- Add CA certs
- Customize the OS layer



Custom Run Image

Use a Dockerfile to customize:

```
FROM docker.io/paketobuildpacks/run-jammy-base

USER root:root

# Install additional packages that we want to make available at run time
RUN apt-get update && \
    apt-get install -y net-tools netcat && \
    rm -rf /var/lib/apt/lists/*

USER cnb:cnb
```

Custom Run Image

Run Image

Layers		
Cmp	Size	Command
109 MB	FROM c0f0e65edcf573a	

Layer Details	
Tags:	(unavailable)
Id:	c0f0e65edcf573ae35e032e0b7f8accec7f0c7763ef4dc82d8f86228c949ebc66.tar
Size:	109 MB
Digest:	sha256:c0f0e65edcf573ae35e032e0b7f8accec7f0c7763ef4dc82d8f86228c949ebc66
Command:	#(nop) USER cnb:cnb

Image Details	
Image name:	dmikusa/my-run-image
Total Image size:	109 MB
Potential wasted space:	0 B
Image efficiency score:	100 %

Count	Total Space	Path
-------	-------------	------

App Image

Layers		
Cmp	Size	Command
109 MB	FROM c0f0e65edcf573a	
11 MB	Layer: 'nginx', Created by buildpack: paketo-buildpacks/nginx@1.0.13	
46 KB	Software Bill-of-Materials	
8.2 KB	Application Layer	
2.8 MB	Buildpacks Application Launcher	
382 B	Buildpacks Launcher Config	
0 B	Buildpacks Process Types	

Layer Details	
Tags:	(unavailable)
Id:	c0f0e65edcf573ae35e032e0b7f8accec7f0c7763ef4dc82d8f86228c949ebc66.tar
Size:	109 MB
Digest:	sha256:c0f0e65edcf573ae35e032e0b7f8accec7f0c7763ef4dc82d8f86228c949ebc66
Command:	

Image Details	
Image name:	apps/nginx
Total Image size:	122 MB
Potential wasted space:	0 B
Image efficiency score:	100 %

Count	Total Space	Path
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^C Quit	Tab Switch view	^F Filter	^L Show layer changes	^A Show aggregated changes
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Levels of Customization

Custom Images

- Totally different build + run images
- A different base OS, distroless or alpine
- Requires creating custom build and run images, plus a custom builder



Levels of Customization

Custom Buildpacks

- Enforce requirements
- Add files or resources to the image
- Set env variables/embed env variables into image
- Custom image labels
- Add CA certificates
- Interact with other buildpacks

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Automating App Builds

CI/CD Support

- **pack** cli runs in some CI/CD systems like GitHub Actions
- **kpack** can build on top of Kubernetes
- CircleCI, Gitlab, Jenkins and Tekton are also supported
- **lifecycle** direct

Enterprise Ready

Paketo Buildpacks Support

- Custom CA certificates
- Proxies for Internet access
- Local mirrors
- Air Gapped Environments





Summary

Buildpacks are...

- Great for Ops Teams
- Great for Dev Teams
- Everyone can be happy

Slides



<https://github.com/dmikusa/buildpacks-for-ops>

Questions?

